

Milestone 1

Specifications

Due: 11:55pm (AU/Malaysia), Friday Week 6

Your project

By now, you have all been assigned a project, which you are going to develop using the SAFe methodology. You should be in the process of being assigned to:

- an agile team within the project
- a speciality group, either
 - Agile Release Train Engineers (RTE)
 - Product Management (PM)
 - Systems Architect (SA)

Everyone is essentially a developer first, with a specialty role on the side, with every team member having a specialty role. This means there will be multiple RTEs, PMs and SAs inside each agile team as well as across agile teams within the project. We recommend 2 RTEs, 2 PMs and 1 SA for a team of 5.

This milestone

This is a short milestone spanning 4 weeks from week 3 to week 6. The usual timeframe for a milestone would be 6 weeks. There will be two important events where you will be coordinating and running the session:

- PI planning meeting (Week 3 workshop)
- End of PI presentation(Week 6 workshop)

The focal point of this milestone would be creating **important process and design documentation prior to actually beginning software development as well as setting up your development environment**. You will be primarily assessed on the deliverables as outlined below as well as individual contribution to each document.

Each project should only have *one set of deliverables* hence you should be working across your groups to facilitate this. All these deliverables are essentially live documents which means they will probably need revision as the project progresses.

In preparation for this milestone, we will be running a planning session during week 3 (of your regular scheduled workshop) for the first Program Increment (PI). If you are unfamiliar with PIs, please brush up on the definitions available in the SAFe framework documentation. There will be a few deliverables that will require some progress even before the planning session so please read carefully.

Before Week 3's PI planning meeting

The Agile Release Train Engineering team is responsible for:

- Developing (and documenting into the PMP) a plan for running the week 3 program increment (PI) planning session, and running the planning session, a template can be found under week 2 on Moodle, we expect this to be heavily modified to fit for your purposes

The Product Management team is responsible for:

- Creating a *Program Backlog* with example user stories (user stories can be fake for now, the idea is to understand the structure) (ClickUp)

The System Architecture team is responsible for:

- Creating a *Potential Solution Architectures* (PSA) document exploring the options of software technologies available to build the solution and communicating them to the project members. This does not include detailed design of every single module; that would be premature. It should only house suggestions.

During Week 3's PI planning meeting and until Week 6

The Agile Release Train Engineering team is responsible for:

- Developing more on the project-wide **Project Management Plan** (PMP) based on SAFe but making clear the specifics relating to your particular solution
- Developing a preliminary version of the **Quality Assurance Plan** (QAP) in your program
- Developing a project-wide **Risk Register** (SAFe doesn't provide much explicit guidance on how to do this, so consider what you've done in past units and consult your lecturers)
- Proposing and implementing a **Work Breakdown Structure** (Google sheet that will track who is responsible for which tasks for each sprint)

The Product Management team is responsible for:

- Developing the **Project Roadmap** (This is a high level document of what features your project aims to achieve at certain points in the year)
- Communicate with the client and relay objectives to rest of the project team
- Developing preliminary solution-level nonfunctional requirements (NFR) as described in the SAFe website and documenting them in the **QAP**
- Ensuring each Agile team has their **Program Backlog** (low level individual tasks) and iteration goals for the first iteration
- Ensuring each Agile team completes their PI Objectives/Backlog Tasks for the first PI

The System Architecture team is responsible for:

- Determining and documenting potential advantages, disadvantages and risks of each architecture in the **Potential Solution Architecture (PSA)** document
- Developing a **Technology Tutorial Plan** which involves creating spikes (a spike is a technical term for the simplest possible program to understand the basics of any technology) for your team be able to understand the basics of React, Meteor, MongoDB or alternatives as well as documenting the execution of said spikes
- Developing a plan for managing the Git Workflow to standardize the processes for working with the GitHub/Gitlab repository for the entire project team (which obviously also relates to quality assurance and hence should be part of your QAP)

Each of these three task sets is going to require consultation with the other specialty teams, so you will be expected to coordinate with them and ensure that your proposals are workable. Furthermore, everyone must participate in the PI planning session in the week 3 class, and you will be assessed on your active contribution to that.

Note that many of these artefacts are preliminary. That means we recognize that they are likely to change over the course of the project – as we all come to understand the project and refine the requirements and the technological and organizational challenges, we will undoubtedly change our minds on some things.

End of PI presentation(Week 6)

At the end of the first PI, students are expected to do a system demo. The demo is a critical aspect of the PI learning cycle for the Agile Release Train (ART). This event

allows stakeholders, customers (or their internal proxies), and senior management to view the progress that the Agile Release Train has made during the PI.

We understand that this is the first PI, and different projects have different levels of complexity in terms of setting up, so we expect very little coding to be done by now.

During the end of PI presentation, the Agile Release Train will present their accomplishments for the entire PI. Clients and stakeholders review the progress in the broader solution context. Think of it as a Sprint Review Meeting for SAFe. Your system demo should draw some inspiration from the description on the Scaled Agile Framework website [here](#).

In the process of completing your first PI, you will have undoubtedly discovered that some things didn't go so well and could be done better. Some of these can be handled at the team level, others will need to be handled at the program (project-wide) level.

Teams are responsible for documenting the outcomes of their team-level retrospectives, including any team-level actions to be taken. It is generally expected that each team will identify at least a couple of team-level improvements - if you can't, the FIT3170 staff will expect a strong justification as to why your process cannot be improved further!

To summarise, please prepare some Google Slides covering the following:

RTEs

- Project background
- Communication channels and management processes, what worked and what didn't (e.g. Sprint duration or meeting times)
- Run through of deliverables and when to use them

PMs

- Present Project Roadmap
- Current State of Program Backlog
- High Fidelity Mockup of Solution

SAs

- Chosen Technology for project (and why alternatives weren't chosen)
- Git management strategies
- The planning and execution of spikes
- Screenshot of commits from multiple members on a fully setup repository

We do not expect any live demonstration of working software as it is early in the semester, however we do expect to see some progress, such as a screenshot of commits to a fully set up Github repository.

Presentation Feedback Form

Everyone needs to fill in a feedback form during the week 6 PI presentations. You will only need to rate other projects, not your own. More details will be included in the form, which will be available on the Moodle Assessments Tab during the presentations.

SAFe Role Peer Assessment Form

We understand that working with a large group can mean that potentially everyone has differences in opinions of ways things are done. The purpose of this unit is to understand how to work together in a team effectively using battle tested processes. An important part of that is understanding who is contributing more and acknowledging that as well as who could use a pick me up in terms of project contribution.

To facilitate this, a compulsory peer assessment form for this and all future milestones where we expect you to rate the contribution of your peers, and this rating will carry weight for your peers. This form also carries weight for being filled out so if you do not complete it, you will lose marks! This milestone we will rate everyone in our Role subteams as majority of the teamwork will be conducted by roles, but next milestone we will be rating those in our Agile Subteams.

These forms will be available under the Moodle Assessments Tab.

What to submit

Outline of deliverables:

- Project Management Plan (PMP)
- Quality Assurance Plan (QAP)
- Risk Register
- Project Roadmap
- Potential Solution Architecture(PSA)
- Technology Tutorial Plan
- Program Backlog(Non document based)

Note: Project team is referring to the big team with up to 15 students, whereas agile subteam is referring to your smaller teams of 4-6 students.

Your entire project team(not agile sub teams) should submit 2 items, comprising:

1. A zip file containing PDF versions of the documents outlined above.
2. A Google Sheet which contains the Work Breakdown Structure.

For non-document based deliverables, you will need to ensure that all markers have access to your shared Google Drive, Clickup board, and any other online system that you are using to organize your project. Failure to share these artifacts with your marker will lose marks. The accounts you must share with are(add only the mentor for your project and our lecturer):

Clayton

- Xiaoning Du, xiaoning.du@monash.edu
- Ian Felix, ian.felix@monash.edu
- Riordan Alfredo, riordan.alfredo1@monash.edu
- Adrian Kristanto, adrian.kristanto@monash.edu
- Zhiming Deng, zhiming.deng@monash.edu
- Baaset Moslih, baaset.moslih@monash.edu

Malaysia

- Shamala Maniam, shamala.maniam@monash.edu

Late submissions

If you do not ask to be marked late, you will be marked on a snapshot of your repositories taken at the due date and time. You may request late marking by emailing the staff.

If you would like an extension, you will find information about the requirements and process at <https://www.monash.edu/exams/changes/special-consideration>. Click on the “In-semester task” tab and open the “Information Technology” section for a link to the correct form.

Marking criteria

When marking your submission, your marker will take into account:

- Correctness of understanding of client’s requirements
- Sufficiency of documentation
- Clarity of documentation

- Presentation – spelling, grammar, readability
- Extent of your contribution to the project

You will receive detailed feedback within two weeks of the due date unless your team is granted an extension. Note that we do not have a page limit of the report, but making it compact and precise is courtesy.

Penalties

If the team requests late marking and has not been granted an extension, a late penalty of 5% per working day will apply.

Although all students in a team will receive the same mark on group tasks, if there is evidence that the distribution of workload in the team is unfair, then we may penalize students who do not appear to be pulling their weight or overly ambitious students who do work assigned to others leaving nothing for them to contribute.

Penalties may also be applied for breaches of the academic honesty policy.